

Everything SATS — Maths Mock 2

Paper 3: reasoning

This is an original practice paper written for Everything SATS. It follows the layout and question types of the real Key Stage 2 mathematics test, but it is **not** a real past paper.

21 questions, 35 marks. Some questions ask you to show your method. You may get a mark for a correct method even if the answer is wrong. You may not use a calculator.

In the real test this paper lasts 40 minutes. There is no need to time yourself here — take the time you need.

For the adult: the answers and how to award marks are in the separate **answer booklet** (Maths Mock 2 answers).

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Name: _____

Paper 3: reasoning

1

Tick **all** the numbers that are less than one million and five thousand.

1,205,000

1,000,600

1,009,000

1,004,000

1 mark

Circle the numbers that have 7 in the **thousands** place.

349,866

717,985

107,039

587,281

156,760

1 mark

2

There were 26,079 runners in a city race.

Round this number to the nearest 1,000.

1 mark

3

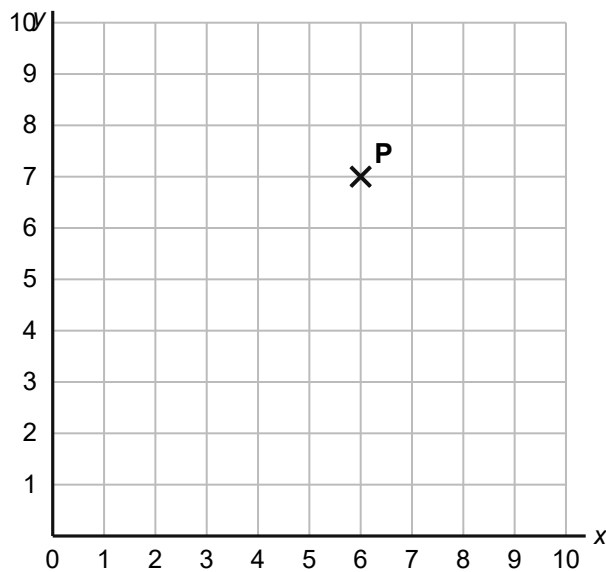
William buys 3 kilograms of carrots and spends £3.51.

What is the cost of one kilogram of carrots?

1 mark

4

Point P is at (6, 7) on the grid.



Point P is translated 4 squares right, 5 squares down and then 3 squares left. What are the coordinates of point P after the translations?

(,)

1 mark

5

The mass of an empty van is 3,125 kilograms. It is loaded with 9 crates. Each crate has a mass of 25 kilograms.

What is the total mass of the van and its load?

Show your method

kg

2 marks

6

Draw lines to match each improper fraction to the equal mixed number.

$$\frac{9}{4}$$

$$2\frac{3}{4}$$

$$\frac{19}{4}$$

$$4\frac{1}{4}$$

$$\frac{17}{4}$$

$$3\frac{3}{4}$$

$$\frac{11}{4}$$

$$2\frac{1}{4}$$

$$4\frac{3}{4}$$

2 marks

7

Ocean buys these three items.

apples
£1.74

cereal
£2.96

bread
£1.03

Ocean pays with a £10 note. How much change does Ocean get?

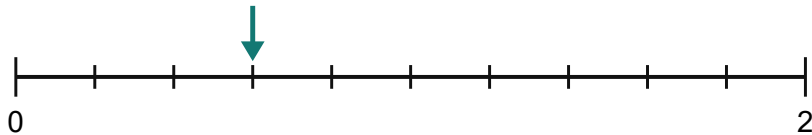
Show your method

£

2 marks

8

Here is part of a number line.



Write the number that the arrow is pointing to.

1 mark

Write the number that is 14 less than 10.

1 mark

9

Here is some information about four animals in a zoo.

	Giraffe	Elephant	Rhino	Hippo
Mass	1,200 kg	5,000 kg	2,100 kg	1,800 kg
Height	5.2 m	3.2 m	1.7 m	1.5 m

The hippo and the rhino together are heavier than the elephant.

The rhino is 20 cm taller than the hippo.

The tallest animal is also the heaviest.

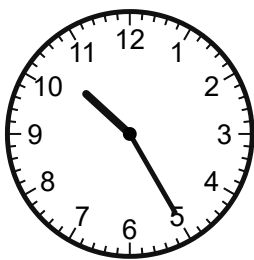
The elephant is more than twice as heavy as the rhino.

Tick the statements that are true.

2 marks

10

This clock shows a time in the afternoon or evening (pm).



Write the time using the 24-hour clock.

1 mark

11

$$\frac{9}{100}$$

0.84

0.538

0.64

Write these numbers in order, starting with the **smallest**.

1 mark

12 Look at this expression: **$y + 6$**

Tick the value of y that makes $y + 6$ a **prime number**.

$y = 4$	☐
$y = 7$	☐
$y = 10$	☐

1 mark

Tick the value of y that makes $y + 6$ a **square number**.

$y = 12$	☐
$y = 16$	☐
$y = 3$	☐

1 mark

13 A factory packs books and comics into boxes.

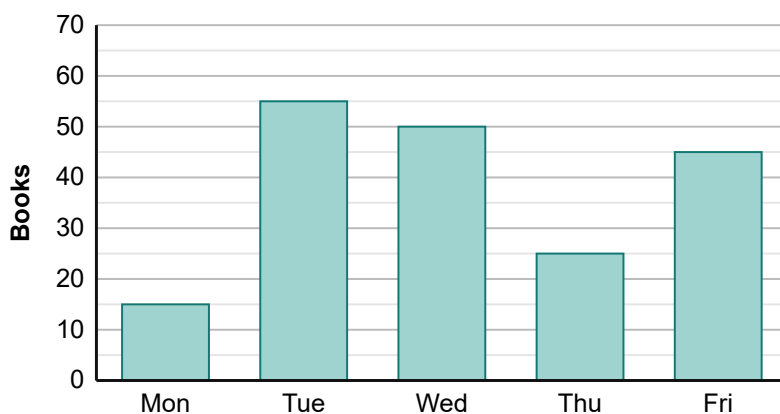
There are 48 books in a box. How many books are there in 95 boxes?

1 mark

There are 40 comics in a box. How many boxes are needed for 4,200 comics?

1 mark

14 This bar chart shows how many books were borrowed from a library each day.



How many more books were borrowed on Wednesday than on Thursday?

1 mark

15

A shop buys 60 boxes of juice cartons. Each box holds 24 cartons. On average, the shop sells 72 cartons each day.

How many days will it take for all of the cartons to be sold?

Show your method

days

3 marks

16

Convert 4.25 km to metres.

m

1 mark

Convert 17.8 cm to metres.

m

1 mark

17

Millie has some wooden blocks that are all the same size, and builds two towers. The taller tower has 8 blocks and is 12 cm tall.

The smaller tower has 6 blocks. What is the height of the smaller tower?

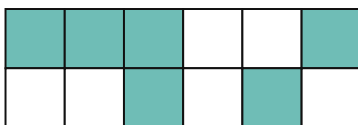
Show your method

cm

2 marks

18

This shape is made from equal squares.



What fraction of the shape is shaded? Give your answer in its simplest form.

1 mark

19

Here is some information about a number:

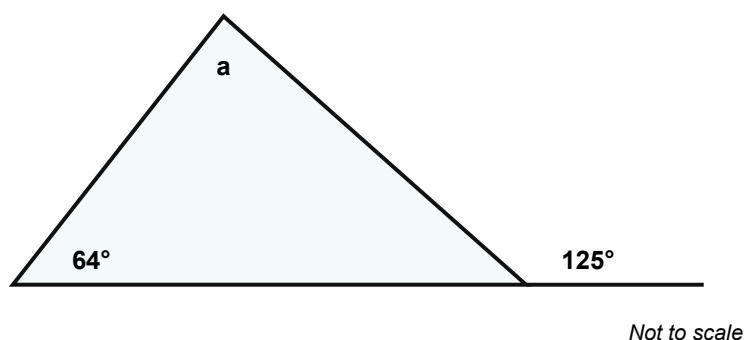
- It has two digits.
- It is a multiple of 8.
- One of its digits is 2.

Write **all** the possible numbers it could be.

2 marks

20

This triangle has one side extended to make a straight line.



Calculate the size of angle a .

Show your method

°

2 marks

21

Tilly scored 20, 37, 15 and 20 points in four games.

What is the mean score?

1 mark